

CSL2050 : Course Project

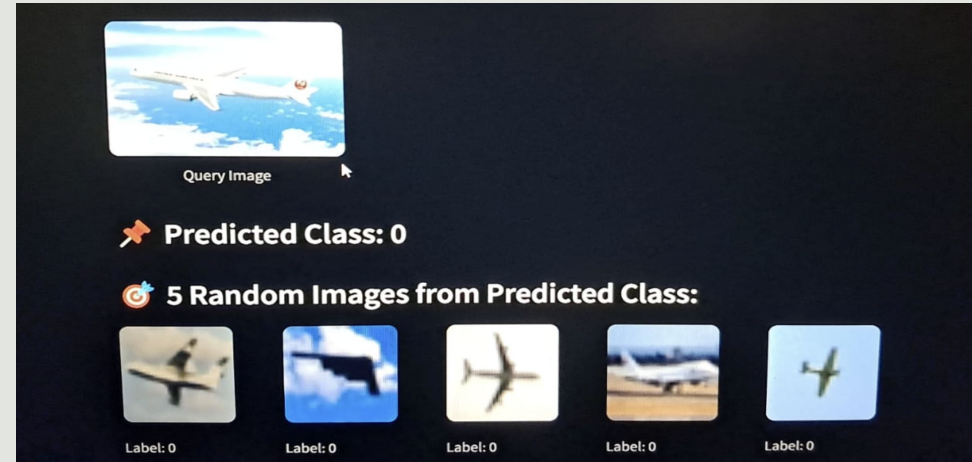
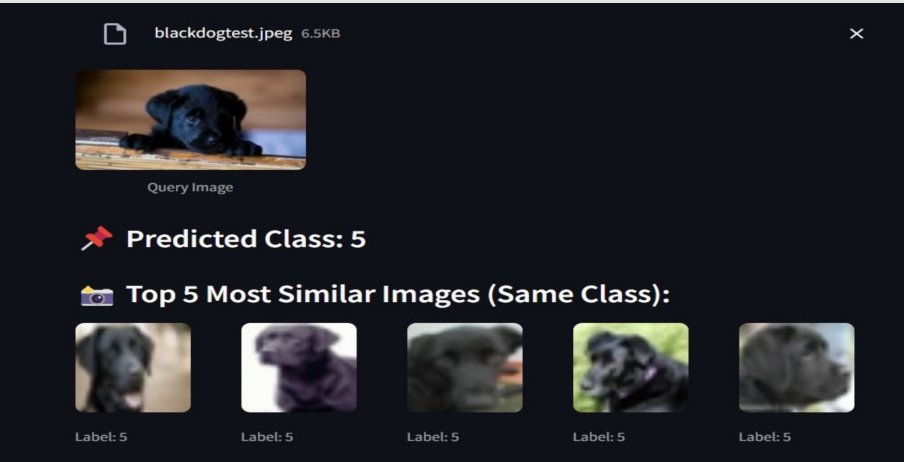
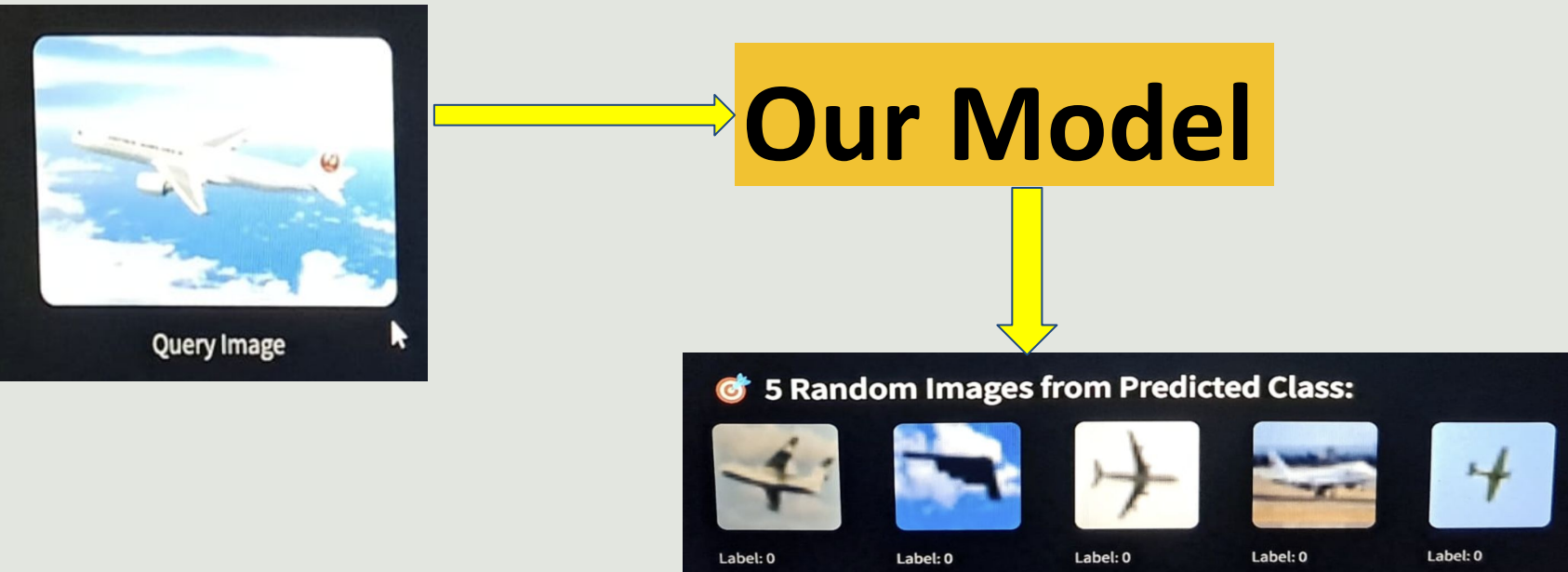


Image Retrieval

Priyanshu, Kalkar Tejas Prasad, S. Vara Prasad Reddy, Kodakandla Manideep,
Neeraj Mansingh, Yadav Karan Subash chandra

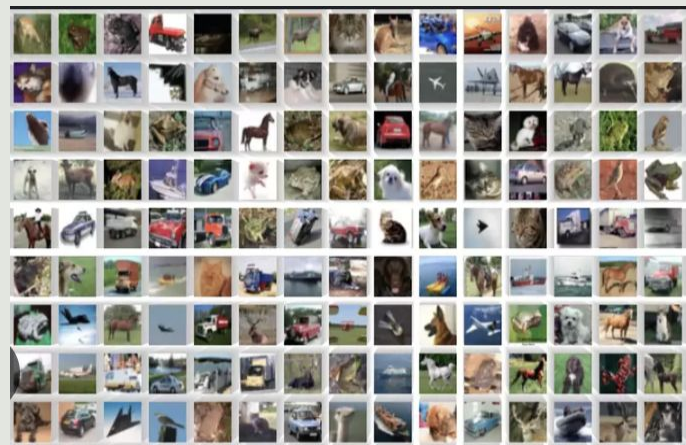
Problem Statement

- (1) In Image Retrieval problem statement we develop algorithms that can effectively search and retrieve images from large databases based on their visual content.
- (2) Our model works by matching images from the CIFAR 10 dataset to the given query images based on the visual features of the query image.



CIFAR 10 Dataset

- (1) **CIFAR-10** is a labeled dataset consisting of **60,000 32x32 color images** in **10 different classes** (like airplanes, cats, trucks, etc.)
- (2) It contains **50,000 training images** and **10,000 test images**, commonly used for image classification tasks.
- (3) The 10 classes are: **airplane**, **automobile**, **bird**, **cat**, **deer**, **dog**, **frog**, **horse**, **ship**, and **truck**.



Preprocessing Techniques

Techniques used

1. PCA (Principal Component Analysis):
 - Reduced dimensionality to preserve 99% variance (654 dimensions).
2. HoG (Histogram of Oriented Gradients):
 - Extracted edge and texture features (324 dimensions).
3. Hybrid Approaches:
 - HoG $\rightarrow$ PCA



Feature Extraction Using CNNs

Models Used:

1. ResNet-50:
 - Pre-trained on ImageNet.
 - Extracted high-level features (2048 dimensions).
 2. QuickNet:
 - Lightweight CNN for efficient feature extraction (512 dimensions).
- Visuals: Network diagrams and feature maps.

Classification Models

- (1) Decision Tree
- (2) Gaussian Naive Bayes
- (3) Gaussian Mixture Model
- (4) K - Nearest Neighbour
- (5) Random Forest
- (6) XGBoost
- (7) Logistic Regression
- (8) Supervised Clustering
- (9) SVM(Support Vector Machines)
- (10) ANN (Artificial Neural Network)

Experimental Results

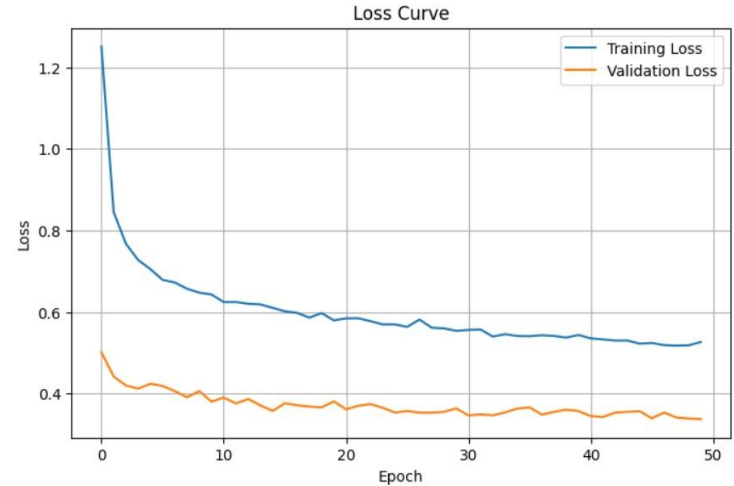
Performance Comparison Table

Features	Best Accuracy	Model
Raw Pixels	38.10%	Logistic Regression
HoG	64.41%	SVM(Radial)
PCA	53.90%	KNN(Bray-Curtis)
HOG+PCA	64.63%	SVM(Radial)
Resnet	86.39%	XGBoost
Resnet	89.76%	ANN (MLP)
Quicknet	75.46%	SVM(Radial)

Experimental Results (ANN)

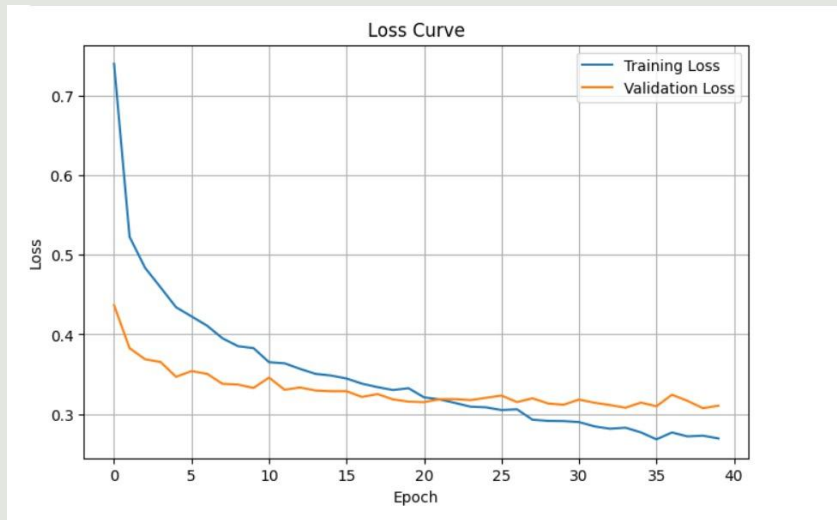


Trial 3, validation loss is above training loss.
Overfitting and confusion seen here

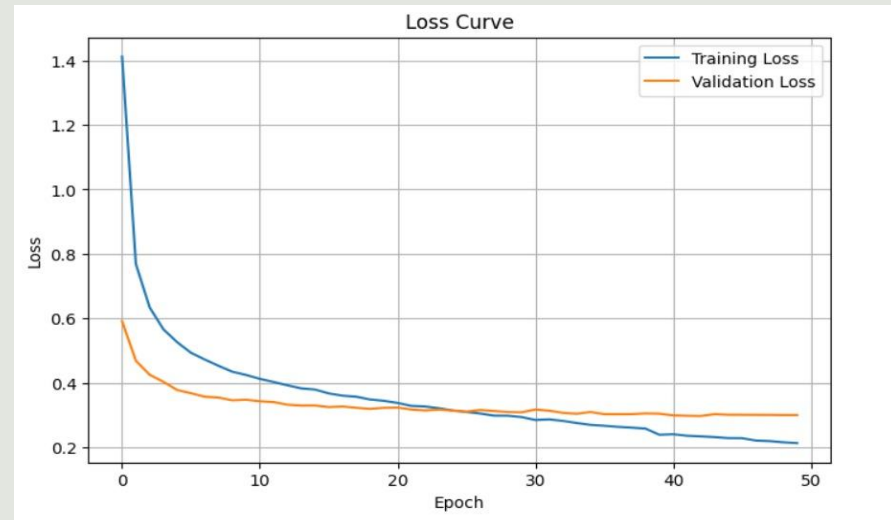


Trial 7, training and val loss do not meet,
underfitting

Experimental Results



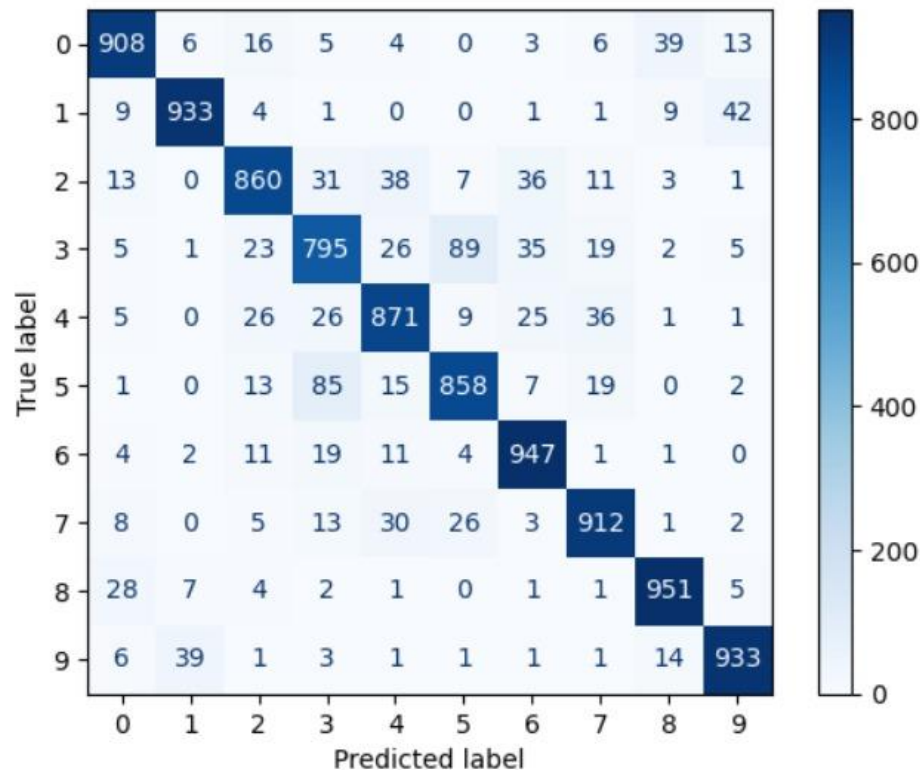
Trial 5, better plot after dropout(0.5) introduced



Final val set used before test set, both losses plots smoothen out due to lr=0.001 in Adams() Optimiser. Also, dropout(0.5) here also.

Experimental Results

Confusion Matrix on Test Set



Here dogs(class 5) and cats(class 3) are getting misclassified mostly.

Key Observations

- (1) ResNet features consistently outperform other feature sets.
- (2) Feature choice became more important than the classifier itself in many cases, especially when comparing raw pixels to processed images.
- (3) QuickNet provides efficient alternatives with competitive accuracy.
- (4) Hybrid approaches like HoG + PCA improve performance but are computationally intensive.
- (5) Raw pixel features perform poorly due to high dimensionality and noise.
- (6) SVM with radial kernel is consistently seen to outperform other models across various features.
- (7) ANN models benefitted from dropout and learning rate scheduling, which helped reduce overfitting and improved generalization.

Conclusion

In our CIFAR-10 experiments, we saw that the type of features used makes a big difference. The best results came from using ResNet features with an MLP, giving an accuracy of 89.48%. Even traditional features like HoG and PCA worked well, especially when combined with models like XGBoost and GMM. XGBoost with HoG-PCA features reached 86.31%, which is quite strong. Overall, deep learning features gave the best performance, but classic models still did a great job when tuned properly.

This shows that combining good feature extraction with the right model can boost accuracy. It also highlights that simpler models can still be competitive if optimized well.

The image features a light gray background with decorative elements in the corners. In the top right and bottom left, there are large, thick blue circular arcs. Along the top right and bottom left edges, there are vertical columns of three small, gold-colored circular icons each.

Thank You